of P and Q. In other words, $P \cong Q$ if and only if P and Q define isomorphic cell complexes (CW-balls) — see Munkres [418] or Björner [89, Sect. 12] [96, Sect. 4.7] for these concepts.

2.3 Polarity

We proceed to construct *polar polytopes*: this is what (nearly) everybody else calls the "dual" of a polytope. We will use the term "polar" in order to distinguish polarity from duality in the sense of (oriented) matroid theory, which we will see later in Lecture 6 in the form of Gale diagrams, the Lawrence construction, and others. (In this, we follow the conventions of [96, pp. 44–45].)

A key observation one should not miss is the *step into dual space* we take in this section. Equivalently, one could just fix a scalar product on $\mathbb{R}^d$, and this way we could construct a polar polytope in the same space as the original. However, there is a lot of choice in any case because the location of the origin is essential for our construction. If we wanted to avoid this, we would have to linearize, and develop "cone polarity" — using some methods of Lecture 1. We will not do this here (in order to keep the principle of making this a lecture on polytopes), but see Exercise 2.13.

Lemma 2.8. *Let P be a polytope in $\mathbb{R}^d$. Then the following conditions are equivalent for $\boldsymbol{y} \in P$.*

(i) *$\boldsymbol{y}$ is not contained in a face of P of dimension smaller than d,*

(ii) *if $\boldsymbol{ay} = a_0$ and $\boldsymbol{a} \neq 0$, then $\boldsymbol{ax} \leq a_0$ is not valid for P,*

(iii) *$\boldsymbol{y}$ can be represented in the form $\boldsymbol{y} = \sum_{i=0}^{d} \lambda_i \boldsymbol{x}_i$ for $d+1$ affinely independent points $\boldsymbol{x}_0, \ldots, \boldsymbol{x}_d \in P$ and for parameters $\lambda_i > 0$ with $\sum_{i=0}^{d} \lambda_i = 1$,*

(iv) *$\boldsymbol{y}$ can be represented as $\boldsymbol{y} = \frac{1}{d+1} \sum_{i=0}^{d} \boldsymbol{x}_i$ for $d+1$ affinely independent points $\boldsymbol{x}_0, \ldots, \boldsymbol{x}_d \in P$.*

Proof. Part (i) implies that P is full-dimensional. From this we get that part (ii) holds: if $\boldsymbol{ax} \leq a_0$ were valid for P, then $\boldsymbol{y}$ would be contained in the face $P \cap \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{ax} = a_0\}$ of smaller dimension. Conversely, if part (ii) holds, then no inequality can define a facet that contains $\boldsymbol{y}$.

Part (iv) trivially implies part (iii), and from this we get part (ii) by an easy calculation:

$$a_0 \;=\; \boldsymbol{ay} \;=\; \sum_{i=0}^{d} \lambda_i \boldsymbol{ax}_i \;\leq\; \sum_{i=0}^{d} \lambda_i a_0 \;=\; a_0.$$

This can hold only if $\boldsymbol{a}\boldsymbol{x}_i = a_0$ holds for all i, so either $\boldsymbol{a} = 0$, or the points $\boldsymbol{x}_i$ are not affinely independent.

Now assume that part (ii) holds. We write P in the form $P = P(A, \boldsymbol{z})$. No equality $\boldsymbol{a}\boldsymbol{x} = a_0$ with $\boldsymbol{a} \neq \boldsymbol{0}$ is valid for P, so P is full-dimensional. We claim that for every $\boldsymbol{u} \in \mathbb{R}^d$ we have $\boldsymbol{y} + \alpha\boldsymbol{u} \in P$, if $\alpha > 0$ is small enough. In fact, this is true unless one of the inequalities that define P is satisfied with equality for $\boldsymbol{y}$, which is excluded by part (ii). But now we can choose $d+1$ possible α-values for $\boldsymbol{u} = \boldsymbol{e}_1, \ldots, \boldsymbol{e}_d, -(\boldsymbol{e}_1 + \ldots + \boldsymbol{e}_d)$, let $\alpha' > 0$ be the minimum of those, and write

$$\boldsymbol{y} \;=\; \frac{1}{d+1}\left\{(\boldsymbol{y} - \alpha'(\boldsymbol{e}_1 + \ldots + \boldsymbol{e}_d)) \;+\; (\boldsymbol{y} + \alpha'\boldsymbol{e}_1) + \ldots + (\boldsymbol{y} + \alpha'\boldsymbol{e}_d)\right\},$$

which is a representation of the desired form. □

If the conditions of Lemma 2.8 are satisfied for $\boldsymbol{y}$, we say that $\boldsymbol{y}$ is an *interior point* of P. Moreover, we use the notation $\mathrm{int}(P)$ for the *interior* of P, which is the set of all interior points of P. (It is easy to verify that this coincides with the usual (topological) definition of the interior of the point set $P \subseteq \mathbb{R}^d$.)

The problem is that the interior of a polytope is not invariant under affine equivalence of polytopes: the center of a triangle is an interior point if the triangle is embedded in $\mathbb{R}^2$, but not if it is embedded in $\mathbb{R}^3$. In fact, $\mathrm{int}(P) = \emptyset$ if P is not full-dimensional in $\mathbb{R}^d$.

Thus, we define the *relative interior* $\mathrm{relint}(P)$ of a polytope, defined as the interior of P with respect to the embedding of P into its affine hull $\mathrm{aff}(P)$, in which P is full-dimensional. Analogous to Lemma 2.8, the following lemma characterizes relative interior points of P.

Lemma 2.9. *Let P be a polytope of dimension $k := \dim(P)$ in some $\mathbb{R}^d$ $(k \leq d)$. Then the following conditions are equivalent for $\boldsymbol{y} \in P$.*

(i) *$\boldsymbol{y}$ is not contained in a proper face of P,*

(ii) *if $\boldsymbol{a}\boldsymbol{x} \leq a_0$ is valid for P, with equality for $\boldsymbol{y}$, then $\boldsymbol{a}\boldsymbol{x} = a_0$ holds for all $\boldsymbol{x} \in P$,*

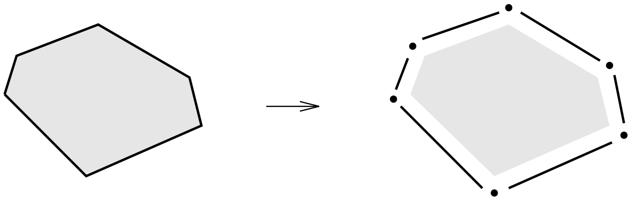
(iii) *$\boldsymbol{y}$ can be represented in the form $\boldsymbol{y} = \sum_{i=0}^{k} \lambda_i \boldsymbol{x}_i$ for $k+1$ affinely independent points $\boldsymbol{x}_0, \ldots, \boldsymbol{x}_k \in P$ and for parameters $\lambda_i > 0$, with $\sum_{i=0}^{k} \lambda_i = 1$,*

(iv) *$\boldsymbol{y}$ can be represented as $\boldsymbol{y} = \frac{1}{k+1}\sum_{i=0}^{k} \boldsymbol{x}_i$ for $k+1$ affinely independent points $\boldsymbol{x}_0, \ldots, \boldsymbol{x}_k \in P$.*

Note that if P is nonempty, then its relative interior contains a point, $\mathrm{relint}(P) \neq \emptyset$. To see this, we can take the barycenter of the vertex set of P, as $\boldsymbol{y} := \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{v}_i$ for $\mathrm{vert}(P) = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_N\}$, or the barycenter of any set of $\dim(P)+1$ affinely independent points (e.g., vertices) in P according to Lemma 2.9(iv).

Furthermore, we get a decomposition of P into a disjoint union of the relative interiors of its faces, from the characterization of Lemma 2.9(i):

$$P \;=\; \biguplus_{F \in L(P)} \operatorname{relint}(F).$$

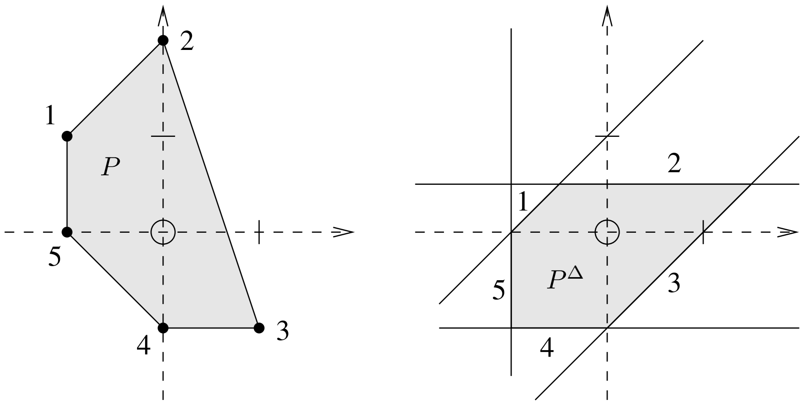


For many problems, in particular for the polarity construction we want to do now, it is convenient to assume that $\mathbf{0} \in \operatorname{int}(P)$ without loss of generality ("w.l.o.g."). This can be achieved, if P is nonempty, by an affine map. In fact, for this we project to $\operatorname{aff}(P)$ and then translate any interior point to $\mathbf{0}$. Now it is easy to see (with Lemma 2.8) that P satisfies this condition if and only if it can be represented as $P = P(A, \mathbf{1})$.

Definition 2.10. For any subset $P \subseteq \mathbb{R}^d$, the *polar set* is defined by

$$P^{\Delta} \;:=\; \{\boldsymbol{c} \in (\mathbb{R}^d)^* : \boldsymbol{c}\boldsymbol{x} \leq 1 \text{ for all } \boldsymbol{x} \in P\} \;\subseteq\; (\mathbb{R}^d)^*.$$

The following shows a convex pentagon P in the plane, determined by its five vertices, and its polar P^{Δ}, a pentagon given by five inequalities.



Clearly, the construction of the polar can be iterated, and thus we get the polar of the polar, or double-polar, as

$$P^{\Delta\Delta} \;=\; \{\boldsymbol{y} \in \mathbb{R}^d : \boldsymbol{c}\boldsymbol{x} \leq 1 \text{ for all } \boldsymbol{x} \in P \text{ implies } \boldsymbol{c}\boldsymbol{y} \leq 1, \text{ for } \boldsymbol{c} \in (\mathbb{R}^d)^*\},$$

where we have identified $\mathbb{R}^d$ and $(\mathbb{R}^d)^{**}$ in the natural way. Now let's examine the nearly obvious (?) basic properties of the polar and double-polar constructions.

Theorem 2.11.

(i) $P \subseteq Q$ *implies* $P^\Delta \supseteq Q^\Delta$ *and* $P^{\Delta\Delta} \subseteq Q^{\Delta\Delta}$,

(ii) $P \subseteq P^{\Delta\Delta}$,

(iii) P^Δ *and* $P^{\Delta\Delta}$ *are convex,*

(iv) $\mathbb{0} \in P^\Delta$, *and* $\mathbf{0} \in P^{\Delta\Delta}$,

(v) *if* P *is a polytope and* $\mathbf{0} \in P$, *then* $P = P^{\Delta\Delta}$,

(vi) *if a polytope* P *with* $\mathbf{0} \in \operatorname{int}(P)$ *is given by* $P = \operatorname{conv}(V)$, *then*

$$P^\Delta \;=\; \{\boldsymbol{a} : \boldsymbol{a}V \leq \mathbb{1}\},$$

(vii) *if a polytope* P *with* $\mathbf{0} \in \operatorname{int}(P)$ *is given by* $P = P(A, \mathbf{1})$, *then*

$$P^\Delta \;=\; \{\boldsymbol{c}A : \boldsymbol{c} \geq \mathbb{0},\ \boldsymbol{c}\mathbf{1} = 1\}.$$

In part (vi), the representation $P = \operatorname{conv}(V)$ means that P is a polytope, and the representation $P = P(A, \mathbf{1})$ in part (vii) implies that $\mathbf{0} \in \operatorname{int}(P)$. Extending the definition of convex hulls and inequality systems to the dual space (i.e., for row vectors), the statements of (vi) and (vii) can be rewritten and combined as

$$P^\Delta \;=\; \operatorname{conv}(A) \;=\; P(V, \mathbb{1}).$$

Proof. Parts (i) to (iv) we can safely leave to the reader: these are routine exercises that should not need note paper.

For part (v), we rely on the conscientious reader to get his or her note pad and do the proof. It also follows from part (vi), for which we compute

$$\begin{aligned} P^\Delta &= \{\boldsymbol{a} : \boldsymbol{a}\boldsymbol{x} \leq 1 \text{ for all } \boldsymbol{x} \in P\} \\ &= \{\boldsymbol{a} : \boldsymbol{a}\boldsymbol{v} \leq 1 \text{ for all } \boldsymbol{v} \in V\}, \end{aligned}$$

where for the last equality "$\subseteq$" is trivial, while "$\supseteq$" follows from convexity (or from a trivial computation).

For part (vii), we compute

$$\begin{aligned} P^\Delta &= \{\boldsymbol{a} : \boldsymbol{a}\boldsymbol{x} \leq 1 \text{ for all } \boldsymbol{x} : A\boldsymbol{x} \leq \mathbf{1}\} \\ &= \{\boldsymbol{c}A : \boldsymbol{c} \geq \mathbb{0},\ \boldsymbol{c}\mathbf{1} = 1\}, \end{aligned}$$

where for the last equality "$\supseteq$" follows from a simple computation $(\boldsymbol{c}A)\boldsymbol{x} = \boldsymbol{c}(A\boldsymbol{x}) \leq \boldsymbol{c}\mathbf{1} = 1$, while "$\subseteq$" is a little harder. For this note that $A\boldsymbol{x} \leq \mathbf{1}$ has a solution. Thus by Farkas lemma III the validity of $\boldsymbol{a}\boldsymbol{x} \leq 1$ implies that there exists a $\boldsymbol{c}' \geq \mathbb{0}$ with $\boldsymbol{c}'A = \boldsymbol{a}$ and $\boldsymbol{c}'\mathbf{1} \leq 1$. Now since P is bounded, there is no $\boldsymbol{x}$ with $A\boldsymbol{x} \leq -\mathbf{1}$, otherwise $\boldsymbol{x} \neq 0$ and $\lambda\boldsymbol{x}$ is in P for all $\lambda \geq 0$.

From this, by Farkas lemma I, there exists a $\boldsymbol{c}'' \geq \mathbb{0}$ with $\boldsymbol{c}''A = \mathbb{0}$ and $\boldsymbol{c}''\mathbf{1} > 0$. With this we can put

$$\boldsymbol{c} := \boldsymbol{c}' + \frac{1 - \boldsymbol{c}'\mathbf{1}}{\boldsymbol{c}''\mathbf{1}}\boldsymbol{c}'',$$

which satisfies $\boldsymbol{c} \geq \mathbb{0}$, $\boldsymbol{c}A = \boldsymbol{c}'A = \boldsymbol{a}$, and $\boldsymbol{c}\mathbf{1} = \boldsymbol{c}'\mathbf{1} + (1 - \boldsymbol{c}'\mathbf{1}) = 1$. □

For illustration of polarity, we also refer the reader to the "classical" pairs of the (regular) cube and octahedron, as sketched in Example 0.4, and of the regular dodecahedron and icosahedron — assuming that they are represented in such a way that $\mathbf{0}$ is the center of symmetry. Furthermore, the polar of any simplex is a simplex (Exercise 2.12).

In fact, it turns out that the combinatorial structure of P^Δ is independent of the exact embedding in $\mathbb{R}^d$, as long as we have $\mathbf{0} \in \operatorname{int}(P)$, as we will see from the next theorem.

For this, we assume that $P \subseteq \mathbb{R}^d$ is again a d-polytope with $\mathbf{0}$ in its interior. In this situation we study, for all faces F of P, the subsets of P^Δ of the form

$$\begin{aligned} F^\diamond \ :=\ \{\boldsymbol{c} \in (\mathbb{R}^d)^* :\quad & \boldsymbol{cx} \leq 1 \text{ for all } \boldsymbol{x} \in P,\\ & \boldsymbol{cx} = 1 \text{ for all } \boldsymbol{x} \in F\} \ \subseteq\ (\mathbb{R}^d)^*. \end{aligned}$$

Theorem 2.12. *Assume that $P = \operatorname{conv}(V) = P(A, \mathbf{1})$ is a polytope in $\mathbb{R}^d$, and that*

$$F = \operatorname{conv}(V') \ =\ \{\boldsymbol{x} \in \mathbb{R}^d : A''\boldsymbol{x} \leq \mathbf{1},\ A'\boldsymbol{x} = \mathbf{1}\}$$

is a face of P, with $V = V' \uplus V''$ and $A = A' \uplus A''$.
(Here we need that all the inequalities $\boldsymbol{ax} \leq 1$ in the system "$A\boldsymbol{x} \leq \mathbf{1}$" that satisfy $F \subseteq \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{ax} = 1\}$ are included in "$A'\boldsymbol{x} \leq \mathbf{1}$.")
Then

$$F^\diamond \ =\ \{\boldsymbol{c}'A' : \boldsymbol{c}' \geq \mathbb{0},\ \boldsymbol{c}'\mathbf{1} = 1\} \ =\ \{\boldsymbol{a} : \boldsymbol{a}V'' \leq \mathbb{1},\ \boldsymbol{a}V' = \mathbb{1}\}.$$

Proof. We compute

$$\begin{aligned} F^\diamond &= \{\boldsymbol{a} : \boldsymbol{ax} \leq 1 \text{ for all } \boldsymbol{x} \in P,\ \boldsymbol{ax} = 1 \text{ for all } \boldsymbol{x} \in F\}\\ &= \{\boldsymbol{a} : \boldsymbol{av} \leq 1 \text{ for all } \boldsymbol{v} \in V,\ \boldsymbol{av} = 1 \text{ for all } \boldsymbol{v} \in V'\}\\ &= \{\boldsymbol{a} : \boldsymbol{av} \leq 1 \text{ for all } \boldsymbol{v} \in V'',\ \boldsymbol{av} = 1 \text{ for all } \boldsymbol{v} \in V'\}\\ &= \{\boldsymbol{a} : \boldsymbol{a}V'' \leq \mathbb{1},\ \boldsymbol{a}V' = \mathbb{1}\}. \end{aligned}$$

For the other half, we use the description of P^Δ in parts (vi) and (vii) of Theorem 2.11, and get

$$\begin{aligned} F^\diamond &= \{\boldsymbol{a} : \boldsymbol{ax} \leq 1 \text{ for all } \boldsymbol{x} \in P,\ \boldsymbol{ax} = 1 \text{ for all } \boldsymbol{x} \in F\}\\ &= \{\boldsymbol{c}A : \boldsymbol{c} \geq \mathbb{0},\ \boldsymbol{c}\mathbf{1} = 1,\ \boldsymbol{c}A\boldsymbol{x} = 1 \text{ for all } \boldsymbol{x} \in F\}\\ &= \{\boldsymbol{c}'A' : \boldsymbol{c}' \geq \mathbb{0},\ \boldsymbol{c}'\mathbf{1} = 1\}, \end{aligned}$$

where for the last equality, "$\supseteq$" is clear, while for "$\subseteq$" we have to work. In fact, for this we can choose some $\boldsymbol{x} \in \operatorname{relint}(F)$, which satisfies $A'\boldsymbol{x} = \mathbf{1}$ and $A''\boldsymbol{x} < \mathbf{1}$, and rewrite $\boldsymbol{c}A = \boldsymbol{c}'A' + \boldsymbol{c}''A''$. Then by

$$1 = (\boldsymbol{c}A)\boldsymbol{x} = (\boldsymbol{c}'A' + \boldsymbol{c}''A'')\boldsymbol{x} = \boldsymbol{c}'(A'\boldsymbol{x}) + \boldsymbol{c}''(A''\boldsymbol{x}) \leq \boldsymbol{c}'\mathbf{1} + \boldsymbol{c}''\mathbf{1} = \boldsymbol{c}\mathbf{1} = 1$$

we have $\boldsymbol{c}''(A''\boldsymbol{x}) = \boldsymbol{c}''\mathbf{1}$, which by $A''\boldsymbol{x} < \mathbf{1}$ implies $\boldsymbol{c}'' = \mathbb{O}$. □

Corollary 2.13. *Let P be a polytope with $\mathbf{0} \in \operatorname{int}(P)$, and let $F, G \in L(P)$ be faces of P. Then*

(i) *$F^\diamond$ is a face of P^Δ,*

(ii) *$F^{\diamond\diamond} = F$, and*

(iii) *$F \subseteq G$ holds if and only if $F^\diamond \supseteq G^\diamond$.*

Corollary 2.14. *The face lattice of P^Δ is the opposite of the face lattice of P:*

$$L(P^\Delta) \cong L(P)^{op}.$$

This, in particular, completes the proof of Theorem 2.7, the last two parts of which we had deferred (remember?). It means that for every statement about the combinatorial structure of polytopes, there is a "polar statement," where the translation reverses inclusion of faces, and interchanges

$$\begin{array}{rcll} \emptyset = \hat{0} & \longleftrightarrow & \hat{1} = P & \\ \text{vertices} & \longleftrightarrow & \text{facets} & \\ \text{edges} & \longleftrightarrow & \text{ridges} & \\ \dots & \longleftrightarrow & \dots, & \quad \text{etc.} \end{array}$$

Note that polarity also identifies the face lattices of facets with (the opposites of) the face lattices of vertex figures. Finally, it says that the "polar" combinatorial descriptions of polytopes, as $\mathcal{V}$-polytopes in terms of vertices, and as $\mathcal{H}$-polytopes in terms of facets, are logically equivalent.

Nevertheless, the metric properties of the polarity construction depend on the location of $\mathbf{0}$ in P, whereas the combinatorial ones do not. This motivates to define that two polytopes P and Q are *combinatorially polar* if $L(P) \cong L(Q)^{op}$. Thus the construction of P^Δ establishes the existence of a combinatorially polar polytope for every polytope P.

2.4 The Representation Theorem for Polytopes

This section has a (by now) simple task: to state and prove the general representation theorem for polytopes. There is no real work left to do: we have assembled all the ingredients, notably Fourier-Motzkin elimination, the Farkas lemmas, Carathéodory's theorem, and polarity. One new term appears in its statement: the *k-skeleton* of a polytope is the union of its k-dimensional faces.